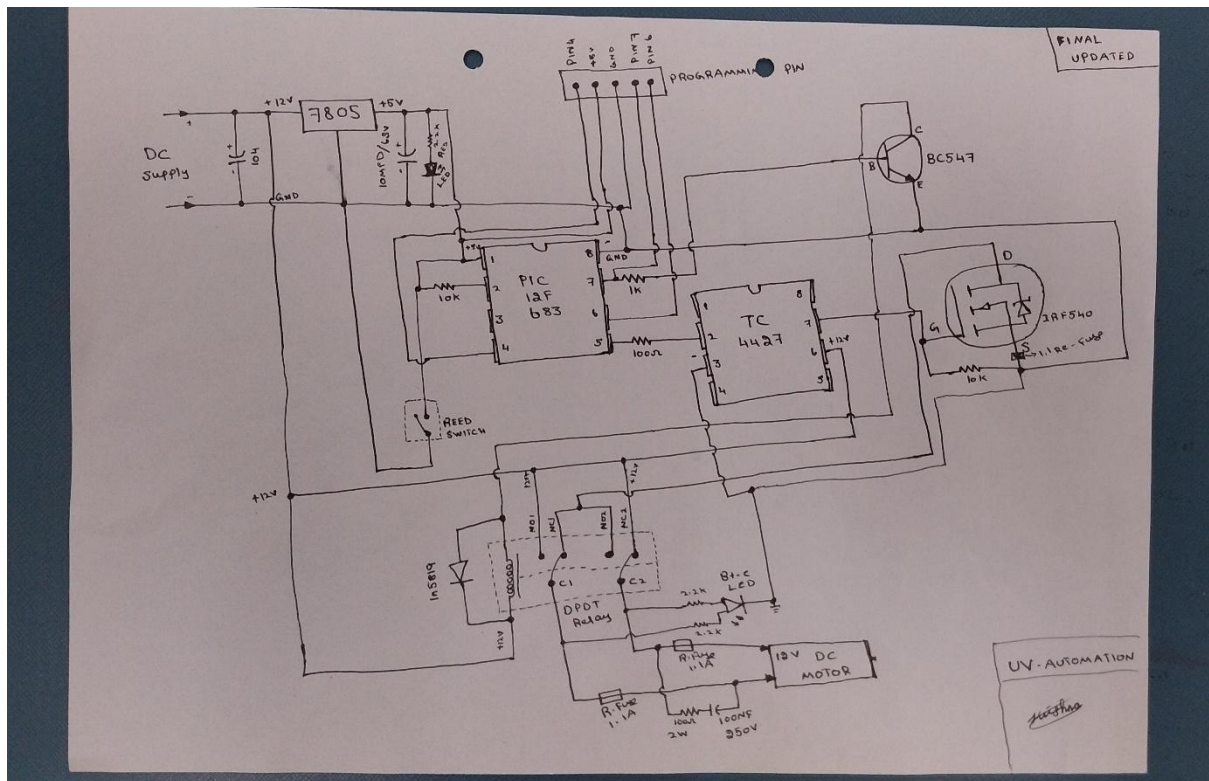




PIC12F683 Pin Details:

Pin 1 - +5v

Pin 8 - Gnd

Pin 2 for Reed switch with 10k resistor pull up

Pin 4 for BC547 Transistor with 1k resistor

Pin 5 for TC4427 IC IN at Pin 2 with resistor 100ohms

TC4427 IC Pin Details:

Pin 2 - IN

Pin 7 - Out (to Mosfet)

pin 3 - Gnd

Pin 6 - +12V

Now here the process is given below:

When i turn On the Circuit it will start the process

Step 1 - Down Cycle the mosfet is triggered by TC4427 for motor running through relay for 37sec.

step 2 - stop the motor for 2sec for rest delay

step 3 - Start Up Cycle, the mosfet and bc547 will turn on at the same time for reversing the polarity.

step 4 - Stop the motor when the reed switch reach the top.

step 5 - rest for 1mins and after 1min the cycle will run again and again.

process step by step :

1. Power On the circuit - it will automatically re-position the brush by use of reed switch. run the motor up for re-position until reed switch detect.

2. if reed switch not detected the motor will run still for 40secs. that 40secs is include that auto re-position. after that 40secs up even reed not detects then stop the motor permently.

3. if all works good and also reed detects, start the down cycle for 37secs.

4. After that 37secs of down cycle, the motor will start up cycle until the reed switch detects. if the reed switch detecets stop the motor and rest for 60secs.

5. After that 60secs rest, again start the the step 3 (down cycle).

6. now test that reed switch. because the reed switch is always low at rest time. so after rest time down the motor for cumpulsary for 10sec. if the reed is still low stop the motor permently, if the reed not detects now, then continuoe the down cycle. and mainly that 10secs also include in that 37secs.

7. after reach 37secs, then start the up cycle and continue this up and down cycle and also test too.

Test Logic Code:

```
#include <xc.h>

#define _XTAL_FREQ 4000000

// CONFIG

#pragma config FOSC = INTOSCIO
#pragma config WDTE = OFF
#pragma config PWRTE = ON
#pragma config MCLRE = OFF
#pragma config BOREN = ON
#pragma config CP = OFF
#pragma config CPD = OFF

// === Pin Definitions ===

#define RELAY      GP0
#define RELAY_TRIS TRISIO0

#define MOTOR_CTRL GP2
#define MOTOR_TRIS TRISIO2

#define REED_SWITCH GP5
#define REED_TRIS   TRISIO5

void init_pins() {

    ANSEL = 0x00;

    CMCON0 = 0x07;
```

```
RELAY_TRIS = 0;
```

```
MOTOR_TRIS = 0;
```

```
REED_TRIS = 1;
```

```
OPTION_REGbits.nGPPU = 0; // Enable global pull-ups
```

```
WPU5 = 1;           // Enable pull-up on GP5 (Reed switch input)
```

```
RELAY = 0;
```

```
MOTOR_CTRL = 0;
```

```
}
```

```
// Debounced reed read
```

```
char read_reed_switch() {
```

```
    if (!REED_SWITCH) {
```

```
        __delay_ms(30);
```

```
        if (!REED_SWITCH)
```

```
            return 1; // LOW = triggered
```

```
    }
```

```
    return 0; // HIGH = not triggered
```

```
}
```

```
void delay_s(unsigned int seconds) {
```

```
    while (seconds--) {
```

```
        __delay_ms(1000);
```

```
    }
```

```
}
```

```
void lockout() {
```

```
MOTOR_CTRL = 0;

__delay_ms(100);

RELAY = 0;

__delay_ms(100);

MOTOR_CTRL = 0;

RELAY = 0;

while (1); // Permanent stop
}
```

```
void down_cycle() {

    RELAY = 0;

    __delay_ms(300);

    MOTOR_CTRL = 1;

    delay_s(37);

    MOTOR_CTRL = 0;

}
```

```
void down_cycle_partial(unsigned int seconds) {

    RELAY = 0;

    __delay_ms(300);

    MOTOR_CTRL = 1;

    delay_s(seconds);

    MOTOR_CTRL = 0;

}
```

```
char down_cycle_partial_check_reed(unsigned int max_seconds) {

    RELAY = 0;

    __delay_ms(300);
```

```
MOTOR_CTRL = 1;
```

```
unsigned int seconds = 0;
```

```
while (seconds < max_seconds) {
```

```
    if (!read_reed_switch()) {
```

```
        MOTOR_CTRL = 0;
```

```
        return 0; // Reed released = safe
```

```
    }
```

```
    __delay_ms(1000);
```

```
    seconds++;
```

```
}
```

```
MOTOR_CTRL = 0;
```

```
return 1; // Still stuck
```

```
}
```

```
void up_cycle() {
```

```
    RELAY = 1;
```

```
    __delay_ms(300);
```

```
    MOTOR_CTRL = 1;
```

```
unsigned int seconds = 0;
```

```
while (!read_reed_switch() && seconds < 42) {
```

```
    __delay_ms(1000);
```

```
    seconds++;
```

```
}
```

```
MOTOR_CTRL = 0;
```

```

    RELAY = 0;
}

char reed_stuck_low_for_seconds(unsigned int seconds) {
    unsigned int count = 0;
    while (count < seconds) {
        if (read_reed_switch()) {
            count++;
        } else {
            return 0; // Released
        }
        __delay_ms(1000);
    }
    return 1; // Fully stuck
}

```

```

void main() {
    init_pins();
    __delay_ms(1000); // Power-on delay

    // === Auto Reposition
    if (!read_reed_switch()) {
        RELAY = 1;
        __delay_ms(300);
        MOTOR_CTRL = 1;

        unsigned int seconds = 0;
        while (!read_reed_switch() && seconds < 38) {

```

```

    __delay_ms(1000);

    seconds++;
}

MOTOR_CTRL = 0;
RELAY = 0;

if (seconds >= 40) {
    lockout(); // Reposition failure
}
}

// === Main Loop
char first_run = 1;

while (1) {
    down_cycle(); // Full 37s down
    up_cycle(); // Until reed triggers or timeout
    delay_s(600); // Rest at top

    if (!first_run) {
        if (reed_stuck_low_for_seconds(60)) {
            if (down_cycle_partial_check_reed(10)) {
                lockout(); // Still stuck after test
            } else {
                down_cycle_partial(27);
                up_cycle();
                delay_s(600);
            }
        }
    }
}

```

```
}
```

```
}
```

```
first_run = 0;
```

```
}
```

```
}
```